WASP-52b

R-0537

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_240908` · created 2026-07-21T22:38:12+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460561.836 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.201 +/- 0.012
Transit depth (Rp/Rs)^2	4.05 +/- 0.47 [%]
Orbital Inclination (inc)	88.78 +/- 1.08
Airmass coefficient 1 (a1)	1.089 +/- 0.018
Airmass coefficient 2 (a2)	-0.075 +/- 0.049
Scatter in the residuals of the lightcurve fit is	1.64 %
Best Comparison Star	#1 - [419, 196]
Optimal Aperture	3.45
Optimal Annulus	21.17
Transit Duration (day)	0.0897 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.201 ± 0.012 vs 0.1646 (deviation 3.03σ)
Duration fit vs published	0.0897 d vs 0.0758 d (deviation 7.3σ)
Mid-time O–C	1.2 min
Depth SNR	16.8
Residual scatter	1.64 %

Light curve

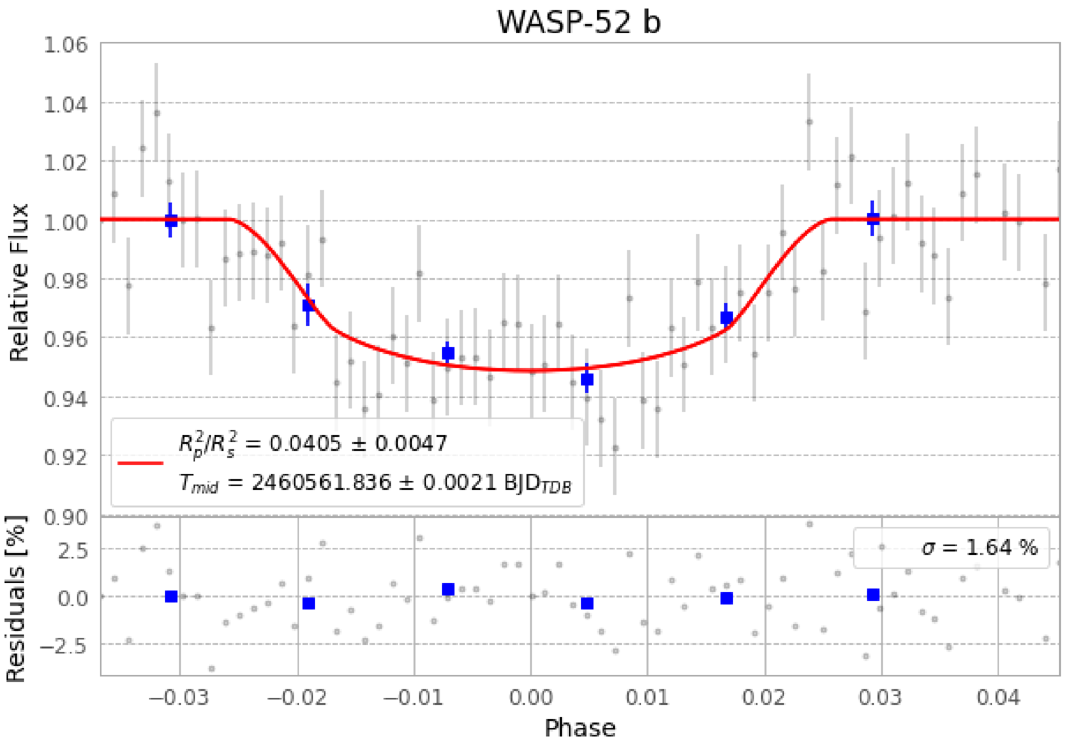

Figure 1 — FinalLightCurve_WASP-52 b_2024-09-07.png (SHA-256 cb1813019cbbbb23…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 216], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 97bd068357c413e6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

70 FITS frames (46 MB), `WASP-52240908062116.FITS` → `WASP-52240908094815.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-240908.log (8ce2d6fb2872…), FinalLightCurve_WASP-52 b_2024-09-07.png (cb1813019cbb…), FinalLightCurve_WASP-52 b_2024-09-07.csv (4489e2ec2566…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)